

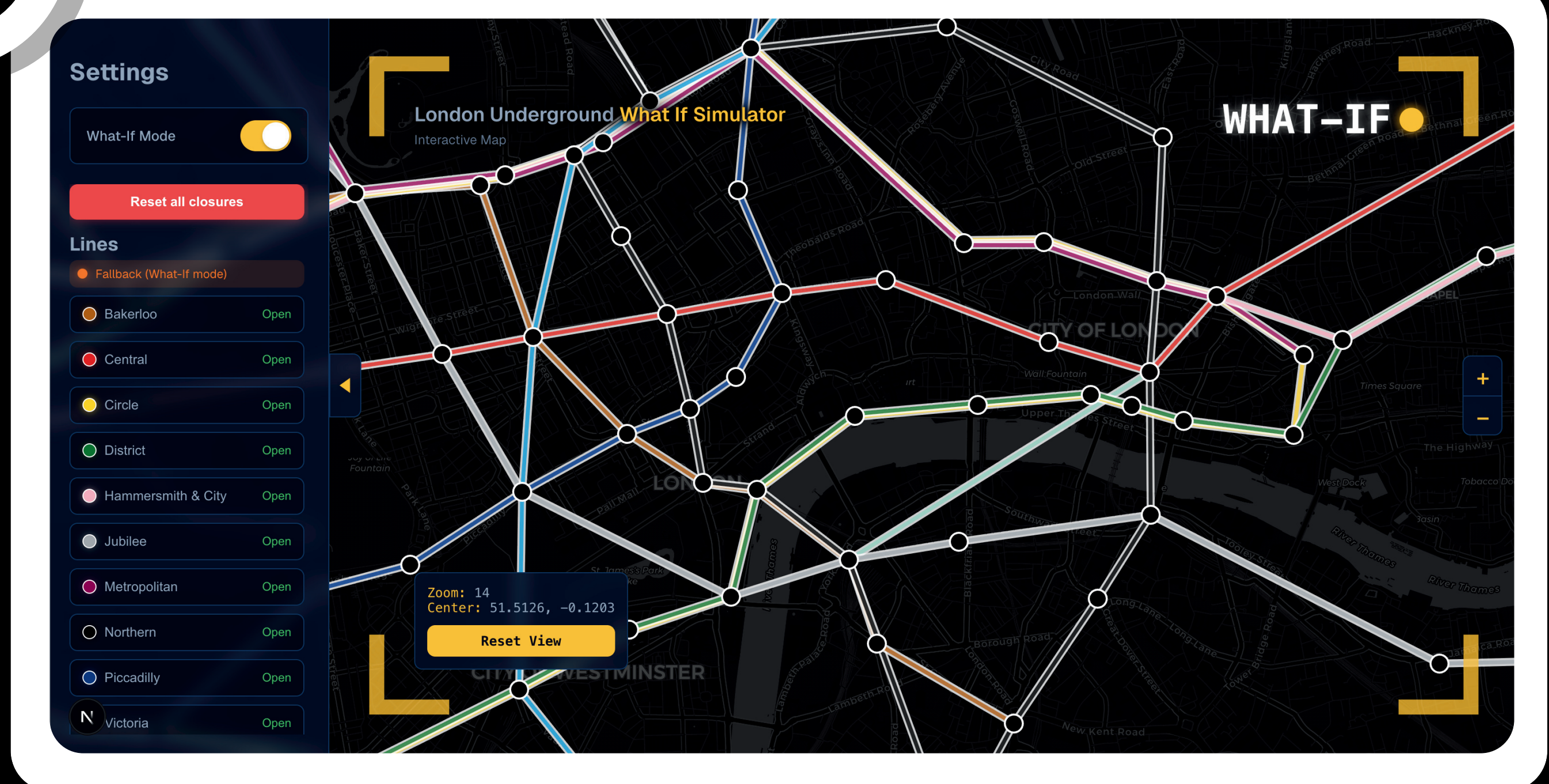
CHALLENGE:

Visualise and simulate the impact of unexpected hypothetical London Underground closures alongside real-time transit conditions.

SOLUTION:

A web-based interactive visualiser and simulator built with Next.js, Neo4j and Leaflet.

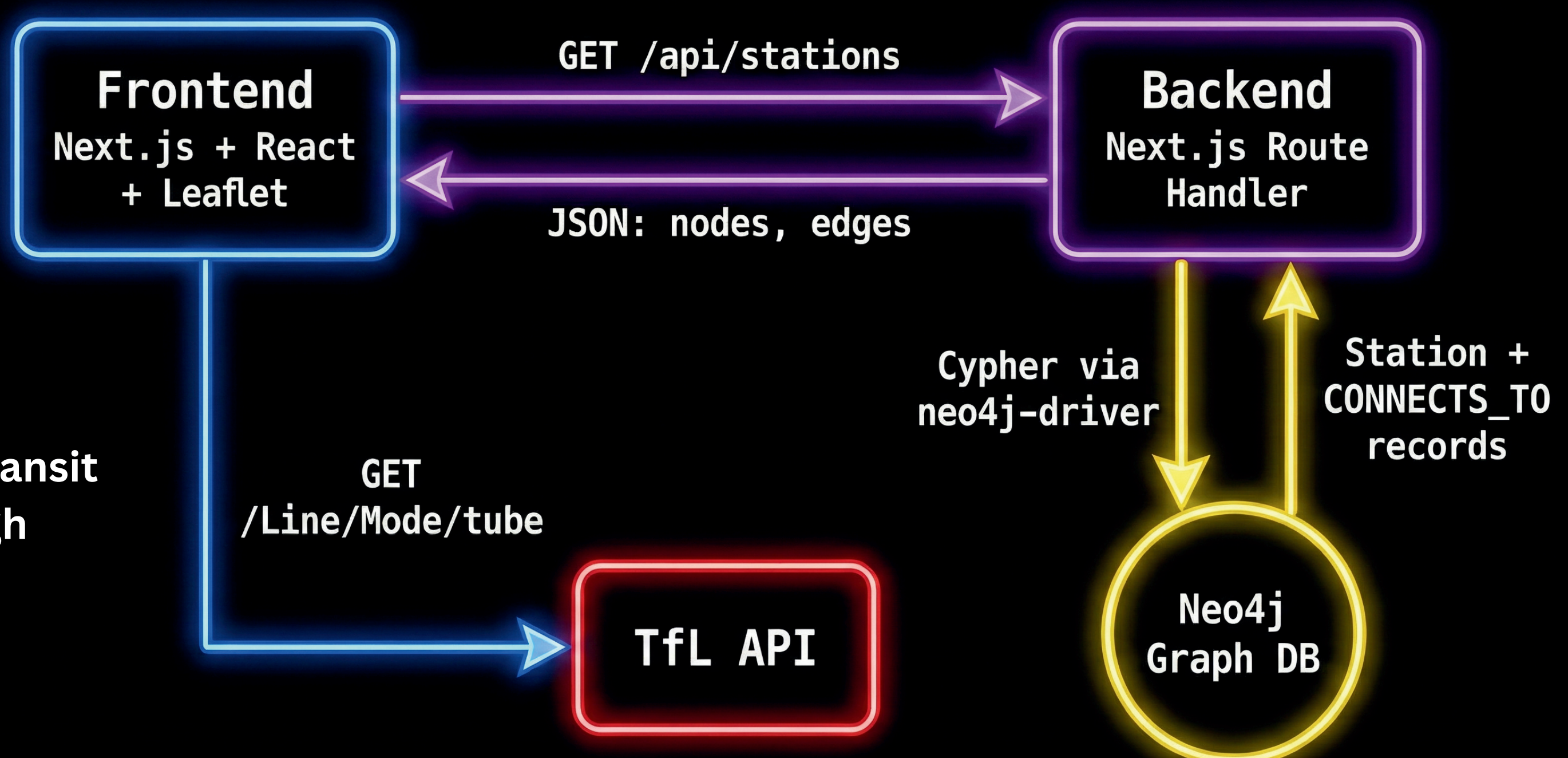
TFL WHAT IF? SIMULATOR



KEY FEATURES:

- **Visualises** real-time train movements across the network using live TfL data feeds.
- **Simulates** targeted station and line closures by letting users manually close graph connections.
- **Recalculates** shortest-path routes in real-time via Dijkstra's algorithm to bypass user disruptions.
- **Delivers a comprehensive tool** to analyse transit resilience and journey predictability through immediate, interactive visual feedback.

SYSTEM ARCHITECTURE:



TECHNOLOGIES USED:



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